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Device and method for producing high-concentration, low-temperature nitric oxide

Abstract

A device and method for forming NO-containing gas flow to treat a biological object is disclosed. The device may include an anode, a cathode, an interelectrode area between the cathode and the anode, an NO-containing gas flow outlet channel leading from the interelectrode area to a nozzle for directing and releasing the NO-containing gas flow from the device and a mechanism to adjust a relative position between the anode and the cathode to produce varying concentrations of NO. In addition, the device may include one or more features for interconnecting the various components to ensure proper and consistent assembly of the device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. Nonprovisional patent application Ser. No. 17/106,311, filed Nov. 30, 2020, which is a continuation of U.S. Non-provisional patent application Ser. No. 15/841,228, filed Dec. 13, 2017, now U.S. Pat. No. 10,850,250, which is non-provisional of U.S. Provisional Patent Application Ser. No. 62/434,018, filed Dec. 14, 2016, titled "A Device and Method for Producing High-Concentration, Low Temperature Nitric Oxide," the entirety of which application is incorporated by reference herein.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to medical equipment and more specifically to devices and methods for providing treatment of a biological object with mixed gases containing nitric oxide. The disclosed methods and devices may be suitable for treating various pathological processes in general, including abdominal, thoracic, purulent, vascular and anaplastic surgery, oncology, urology, combustiology, dentistry, ophthalmology, podiatry, neurosurgery and other fields of medicine.

BACKGROUND OF THE DISCLOSURE

(2) A device and method for preparing a mixture containing air and nitrogen for treating medical pathologies by directly supplying the mixture to a patient's tissue is known. For example, U.S. Pat. No. 7,498,000, the entire contents of which are incorporated by reference, discloses a device and method for forming a nitric oxide (NO)-containing gas flow to treat a biologic object.

(3) NO is a pharmaceutical molecule that generates a dose related response when used to treat a biologic object. Therefore, NO must be delivered in a dosage or concentration that achieves a desired treatment effect. Due to the complexities of biological systems and tissues, NO concentrations levels required to achieve a desired effect will differ based on the application. High NO concentration levels required for one treatment may be toxic if used in a different treatment. For medical and veterinary applications, it is essential for the device to have the ability to produce NO at various concentration levels. One limitation of known devices and methods is that the concentration of NO generated is limited by the device having a fixed distance between the anode and the cathode, and matching fixed electrical parameters of the discharge.

(4) For effective treatment, it is also important for the concentration of NO produced from the device to be consistent.

(5) In addition, repeatable and consistent manufacturability of the device is important, but difficult to achieve. For a NO producing device to operate effectively, it is important to maintain a consistent and repeatable arc gap to control the output parameters. However, precise manufacturing of a device made from a number of interconnected components can be a challenge.

(6) In view of the foregoing, it would be desirable to provide an improved device and method that overcomes the deficiencies and limitations associated with the prior art device. In addition, it would be desirable to provide an improved device that is easier to manufacture.

SUMMARY OF THE DISCLOSURE

(7) This Summary is provided to introduce a selection of concepts in a simplified form that are

further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended as an aid in determining the scope of the claimed subject matter.

(8) Disclosed herein is an improved device for forming NO-containing plasma gas flow to treat a biological object. In one embodiment, the device may include an anode, a cathode, an interelectrode area between the cathode and the anode, an NO-containing gas flow outlet channel leading from the interelectrode area to a nozzle for directing and releasing the NO-containing gas flow from the device, and means to adjust a relative position between the anode and the cathode to produce varying concentrations of NO.

(9) In one embodiment, the device may include a plurality of holes formed in an outer perimeter of a distal end of the cathode, the holes creating a vortex of airflow as the gas flow passes through the interelectrode area.

(10) In one embodiment, the device may include a coolant pathway coupled to a fluid entry port and a fluid exit port so that a cooling fluid can be introduced into the device via the fluid entry port, circulated thru the device via the coolant pathway prior to exiting through the fluid exit port. The cooling fluid may be dielectric. Alternatively, the cooling fluid may be electrically conductive.

(11) In one embodiment, the device may further include a coolant divider for separating the NO-containing gas flow outlet channel from the cooling fluid while directing the NO-containing gas to the nozzle.

(12) In one embodiment, the device may further include an outer housing, the anode being electrically coupled to the outer housing of the device. The device may also include an insulator positioned between a portion of the cathode and a portion of the outer housing of the device to insulate the portion of the cathode from the outer housing of the device. The device may also include an interim electrode coupled to the insulator, the interim electrode positioned adjacent to a hafnium tip associated with the cathode so that an electrical arc occurs between the interim electrode and the hafnium tip, the electrical arc emanating into the interelectrode area.

(13) In one embodiment, the outer housing of the device may include a lower shell and an upper shell. The lower shell may include a plurality of threads for engaging a corresponding plurality of threads formed on the upper shell. In one embodiment, the cathode may be located within the lower shell, the insulator being positioned between the cathode and the lower shell to insulate the cathode and the lower shell from one another.

(14) In one embodiment, the lower shell may include a first lower coolant pathway coupled to a fluid entry port and a second lower coolant pathway coupled to a fluid exit port. The upper shell may include a first upper coolant pathway in fluid communication with the first lower coolant pathway and a second upper coolant pathway in fluid communication with the second lower coolant pathway so that, when the lower shell is coupled to the upper shell, the first and second lower coolant pathways are aligned with the first and second upper coolant pathways.

(15) In one embodiment, the insulator may include one or more coolant entryways for providing a pathway for the cooling fluid to interact with the cathode. The one or more coolant entryways may provide a pathway for the cooling fluid to directly contact an upper portion of the cathode prior to the interelectrode area.

(16) In one embodiment, the nozzle includes a compression nut so that rotation of the nozzle advances the anode towards the cathode, the nut including a plurality of threads for engaging a corresponding plurality of threads formed on an outer housing of the device. In one embodiment, the nozzle may further include a jacket assembly, a nozzle tip driver and a nozzle tip for releasing the NO-containing gas. The jacket assembly may be in the form of the compression nut so that rotation of the nozzle tip driver advances the anode (e.g., the nozzle tip, the coolant divider, and the arc chamber) towards the cathode. The jacket assembly may include a plurality of threads for engaging a corresponding plurality of threads formed on an outer housing of the device. The nozzle tip driver may include a plurality of threads for engaging a corresponding plurality of threads

formed on the jacket assembly.

(17) In one embodiment, the cathode may include a central hollow portion for providing a pathway for air to be forced through the cathode and into the interelectrode area.

(18) In one embodiment, the device may further include an insulator positioned between a portion of the cathode and a portion of an outer housing of the device to insulate the cathode from the outer housing of the device. The device may also include a cathode insulator for providing a dielectric barrier between a portion of the outer housing of the device and a portion of the cathode. The insulator may be coupled to the cathode insulator by a complementary coupling feature.

(19) In one embodiment, the complementary coupling feature may include a recess formed on one of the insulator or cathode insulator for receiving a projection formed on the other one of the insulator or cathode insulator. The cathode insulator may include an internal borehole for receiving a portion of the insulator, the recess and the projection mating with one another when the insulator is received within the internal borehole of the cathode insulator to secure a position of the cathode insulator with respect to the insulator.

(20) In one embodiment, the device may further include an alignment bushing and a complementary coupling feature for coupling the alignment bushing to the cathode insulator. In one embodiment, the complementary coupling feature for coupling the alignment bushing to the cathode insulator may include a recess formed on one of the alignment bushing or cathode insulator for receiving a projection formed on the other one of the alignment bushing or cathode insulator to secure a position of the cathode insulator with respect to the alignment bushing.

(21) In one embodiment, the device may also include a complementary coupling feature for coupling the alignment bushing to the coolant divider. In one embodiment, the complementary coupling feature for coupling the alignment bushing to the coolant divider may include a recess formed on one of the alignment bushing or coolant divider for receiving a projection formed on the other one of the alignment bushing or coolant divider to secure a position of the coolant divider with respect to the alignment bushing.

(22) In one embodiment, the alignment bushing may include an internal borehole for receiving a portion of the coolant divider, the recess and the projection mating with one another when the coolant divider is received within the internal borehole of the alignment bushing to secure a position of the coolant divider with respect to the alignment bushing.

(23) In one embodiment, the device may include an insulator positioned between a portion of the cathode and a portion of an outer housing of the device to insulate the outer housing of the device from the portion of the cathode. In addition, the device may include an interim electrode coupled to the insulator and a chamber standoff disposed between the interim electrode and the interelectrode area. The chamber standoff may be constructed of a dielectric material that prevents electrical connection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) By way of example, specific embodiments of the disclosed device will now be described, with reference to the accompanying drawings, in which:

(2) FIG. 1 is a perspective view illustrating an exemplary embodiment of an NO producing device for providing treatment of a biological object with mixed gases containing nitric oxide;

(3) FIG. 2 is a cross sectional view of the device shown in FIG. 1, taken along line 2-2 of FIG. 1;

(4) FIG. 3 is a second cross sectional view of the device shown in FIG. 1, taken along line 2-2 of FIG. 1;

(5) FIG. 3A is a detailed view illustrating an interelectrode area of the device shown in FIG. 3;

(6) FIG. 4 is an exploded view of the device shown in FIG. 1;

- (7) FIG. 5A is a side view an exemplary embodiment of an insulator and an exemplary embodiment of an interim electrode coupled to the insulator for use in the device shown in FIG. 1;
- (8) FIG. 5B is an exploded view of the insulator and interim electrode shown in FIG. 5A;
- (9) FIG. 5C is a detailed view of the insulator and interim electrode shown in FIG. 5A;
- (10) FIG. 6A is a perspective view of the insulator shown in FIG. 5A;
- (11) FIG. 6B is a perspective view an exemplary embodiment of a cathode insulator for use in the device shown in FIG. 1;
- (12) FIG. 6C is an end view of the cathode insulator shown in FIG. 6B coupled to the insulator shown in FIG. 6A;
- (13) FIG. 7A is a side view an exemplary embodiment of an alignment bushing coupled to the cathode insulator shown in FIG. 6B;
- (14) FIG. 7B is an exploded view of the alignment bushing and cathode insulator shown in FIG. 7A;
- (15) FIG. 8A is an exploded, perspective view an exemplary embodiment of a coolant divider and alignment bushing;
- (16) FIG. 8B is an end view of the alignment bushing coupled to the coolant divider shown in FIG. 8A;
- (17) FIG. 9A is a side view of an exemplary embodiment of a cathode tip that may be used with the device shown in FIG. 1;
- (18) FIG. 9B is a cross-sectional view of the cathode tip shown in FIG. 9A, taken along line 9B-9B in FIG. 9A;
- (19) FIG. 9C is a cross-sectional view of the cathode tip shown in FIG. 9A, taken along line 9C-9C in FIG. 9A;
- (20) FIG. 10A is a side view of an alternate example embodiment of a threadably adjustable anode; and
- (21) FIG. 10B is a cross-sectional view of the adjustable anode shown in FIG. 10A, taken along line 10B-10B in FIG. 10A.

DETAILED DESCRIPTION

(22) A device and method in accordance with the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the device and method are shown. The disclosed device and method, however, may be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the device and method to those skilled in the art. In addition, the disclosed device will be illustrated and described as containing a number of novel features, it should be understood that the device is not limited to containing all of the described features but rather the device may contain one or more of the described features. In the drawings, like numbers refer to like elements throughout.

(23) The present disclosure describes an improved device and method for providing treatment of a biological object with mixed gases containing nitric oxide (referred to herein as NO-containing gas). Referring to FIGS. 1-4, the improved device 10 for producing and delivering NO-containing gas is shown. Generally speaking, the device 10 may form an NO-containing gas for treating a biologic object. The device 10 may form the NO-containing gas by using a DC arc discharge to generate NO in an interelectrode area 500 (e.g., an area between a cathode and an anode). As will be described in greater detail, the device 10 may be configured to produce and release, from a tip of the device 10, various concentrations of nitric oxide ("NO") in more consistent output as compared to current devices. Specifically, the device 10 may, because of one or more specifically engineered confluences of high voltage electrical arc-discharge, high velocity air impingement, and recirculating cooling fluid, produce various concentrations of NO-containing gas as compared to prior art devices so that the improved device allows for selective treatment of a specific biological

object such as, for example, blood coagulation, wound healing, scar removal, disinfection, etc.

(24) Generally speaking, the device **10** includes an anode **20**, a cathode **60**, an interim electrode **120** for producing NO-containing gas, and an NO-containing gas flow outlet channel **152** leading from the interelectrode area **500** to a nozzle **200** for directing and releasing the NO-containing gas flow from the device **10**. In use, the anode **20** may be formed as a single component. Alternatively, the anode **20** may be manufactured from multiple components, which are then coupled together. As illustrated, the anode **20** may include a proximal anode **150**, a coolant divider **160**, and the nozzle **200**. Moreover, as illustrated, the proximal anode **150**, the coolant divider **160**, and the nozzle **200** may be threadably coupled to one another, although it is envisioned that they may be coupled by any other means including, for example, welding, press-fit, etc.

(25) By design, as will be described in greater detail below, the position of the anode **20** with respect to the cathode **60** can be adjusted to produce various concentrations of NO-containing gas. That is, referring to FIG. 3A, according to one aspect of the present disclosure, the anode **20**, and more specifically, the proximal anode **150**, may have a proximal surface **21**. In addition, the cathode **60**, and more specifically, the hafnium tip **80** may have a distal surface **61**. The longitudinal distance X, as measured between the distal end **61** of the cathode **60** and the proximal surface **21** of the anode **20**, can be varied to produce different NO concentrations. As will be understood, varying the offset or distance X between the distal end **61** of the cathode **60** and the proximal surface **21** of the anode **20**, varies the amount of voltage needed to be applied to the cathode **60** to maintain the plasma arc, and thus varies the NO concentration in the gas. In use, the distance X between the distal end **61** of the cathode **60** and the proximal surface **21** of the anode **20** can be adjusted by any suitable mechanism now known or hereafter developed including, but not limited to, those mechanism described herein.

(26) The device **10** may also include an outer housing **30**. The outer housing **30** may comprise the exterior housing or enclosure of the device **10**. In addition, in use and as will be described in greater detail below, the outer housing **30** acts or receives an electrical ground. Moreover, the outer housing **30** acts as a container or housing for containing and directing fluid within the device **10**. In the illustrated embodiment, the outer housing **30** may be formed from two components, e.g., a lower shell **32** and an upper shell **40**, for ease of assembly. As will be described in greater detail below, the lower shell **32** and the upper shell **40** may be threadably coupled to one another, although it is envisioned that they may be coupled by other means including being manufactured as a single component.

(27) The cathode **60** may be formed as a single component. Alternatively, the cathode **60** may be manufactured from multiple components, which are then coupled together. As illustrated, the cathode **60** may include a proximal cathode **62** and a distal cathode **64**. In use, the cathode **60** may be centrally located within the lower shell **32**. As illustrated, the proximal cathode **62** and the distal cathode **64** may be threadably coupled to one another, although it is envisioned that they may be coupled by any other means including, for example, welding, press-fit, etc.

(28) In one embodiment, the cathode **60** may also include a hafnium tip **80** associated with the distal cathode **64**. The hafnium tip **80** may be coupled to the cathode **60** (e.g., distal cathode **64**) by any means including, for example, a press-fit. In use, the hafnium tip **80** acts to provide or release electrons to stabilize or maintain the arc. Alternatively, it is envisioned that metals other than hafnium may also be used. Moreover, in an alternate embodiment, it is envisioned that the cathode may be “tip-less”, for example, for high-voltage/low-current versions of the device. As previously mentioned, the position of the cathode **60** with respect to the anode **20** may be adjusted to produce various concentrations of NO-containing gas. That is, as will be described in greater detail, in one embodiment, the distance between the cathode **60** and the anode **20** may be adjusted to produce different concentrations of NO-containing gas.

(29) The device **10** may also include an insulator **100** positioned between at least a portion of the cathode **60** and at least a portion of the outer housing **30**, for example, the lower shell **32**, to

insulate at least a portion of the outer housing **30** from the cathode **60**. As will be described in greater detail below, an interim electrode **120** may be positioned at a distal end of the insulator **100**. As will be described in greater detail below, the insulator **100** may be coupled via, for example, a threaded coupling to the outer housing **30** (e.g., at a proximal end **34** of the lower shell **32**). In addition, the insulator **100** may be coupled via, for example, a threaded coupling to the cathode **60** (e.g., at proximal end of the proximal cathode **62**). By providing a threaded connection between the cathode **60** (e.g., the proximal cathode **62**) and the insulator **100**, adjustment of the relative position of the cathode **60** (and hence the hafnium tip **80** coupled to a distal end thereof) with respect to the interim electrode **120**, which is coupled to the distal end of the insulator **100**, can be adjustably varied to, for example, vary the NO concentration. As such, in use, the insulator **100** helps fix the spacing between the interim electrode **120** and the cathode **60**. In use, when assembled, the interim electrode **120** may be positioned within or adjacent to the anode **20** (e.g., the proximal anode **150**) and the cathode **60** to provide ignition and stability of the electrical arc between the anode **20** and the cathode **60**.

(30) In use, a high voltage supply can be connected to the cathode **60** while the outer housing **30** may be connected to ground. In addition, the outer housing **30** may be in electrical contact with the anode **20**, for example, the upper shell **40** may be in electrical contact with the nozzle tip **200**, thus enabling the anode **20** to be electrically grounded so that, in use, an arc is generated in the interelectrode area **500** (e.g., area between the cathode **60** and anode **20**). That is, in use, the nozzle tip **200**, the nozzle tip driver **220**, and a jacket assembly **210** associated with the nozzle tip **200** may be in contact with the outer housing **30** and thus electrically grounded. The grounding may be electrically transferred to the coolant divider **160** and the proximal anode **150**, which may also be grounded as a result, so that the arc is generated in the interelectrode area **500** between the cathode **60** and the anode **20**. As previously mentioned, the insulator **100** may be positioned between the cathode **60** and a portion of the outer housing **30** (e.g., between the cathode **60** and a portion of the lower shell **32**) to insulate at least a portion of the outer housing **30** from the cathode **60**. In addition, the anode **20** and the cathode **60** are electrically insulated with respect to the interim electrode **120**, which is located adjacent the hafnium tip **80**. As such, the interim electrode **120** is said to have a “floating” potential. By this arrangement, the electrical arc occurs at the juncture between the interim electrode **120** and the hafnium tip **80**, emanating into the interelectrode area **500**. Upon being generated in the interelectrode area **500**, the NO-containing gas is directed thru the NO-containing gas flow outlet channel **152** to the nozzle tip **200**. Specifically, the NO-containing gas passes thru respective output flow channels formed in the proximal anode **150**, the coolant divider **160** and the nozzle tip **200**, which output flow channels are in line with the cathode **60** to provide the output flow of the NO-containing gas through the nozzle tip **200**.

(31) The device **10** may use any method now known or hereafter developed in order to create the DC arc discharge in the interelectrode area **500** (e.g., area between the cathode **60** and the anode **20**). For example, the device **10** may include a voltage supply for applying a voltage to the cathode **60** to generate and maintain a DC arc discharge between them, wherein a positive potential is applied to the outer housing **30** and a negative potential is applied to the cathode **60**. The proximal anode **150** may be in fluid communication with a pathway **330** (e.g., a central hollow portion) formed in the cathode **60** for injecting a source gas into the interelectrode area **500** (e.g., area between the cathode **60** and the anode **20**), where the source gas contains at least oxygen and nitrogen. The interelectrode area **500** may also be in fluid communication with the NO-containing gas flow outlet channel **152** formed in the anode **20** (e.g., the proximal anode **150**, the coolant divider **160** and the nozzle tip **200**) for directing the NO-containing gas flow from the interelectrode area **500**. The NO-containing gas flow outlet channel **152** may further direct the NO-containing gas flow to the nozzle tip **200** so that the NO-containing gas flow can be used to treat a biologic object. As will be understood, the NO-containing gas flow is formed in the interelectrode area **500** between the anode **20** (e.g., proximal anode **150**), the interim electrode **120**, and the

cathode **60** from a source gas under the effect of a DC arc discharge generated and maintained in the interelectrode area **500** between the anode **20** (e.g., proximal anode **150**) and the cathode **60**.

(32) The arc discharge between the anode **20** and the cathode **60** may be generated by providing an open-circuit DC voltage across the cathode **60** and forming one or a series of high-voltage pulses to generate a spark discharge between the anode **20** and cathode **60**. A value of the open-circuit voltage may be selected and adjusted to provide the change of the spark discharge to a stationary arc discharge. In non-limiting exemplary embodiments, the open-circuit voltage can be at least 200 V, and the high-voltage pulse can be at least 4 kV although other value combinations of open and high pulse voltages may be implemented. Furthermore, the stationary DC arc discharge may be maintained by a current of at least 1.8 A, where the arc discharge is stabilized using the interim electrode **120** to provide steady generation of plasma in the interelectrode area **500** (e.g., area between the cathode **60** and the anode **20**) across the electric arc formed between the cathode **60** and the anode **20**.

(33) Referring again to FIG. **1**, the tip of the device **10** may include a jacket assembly **210**, a nozzle tip driver **220** and the nozzle tip **200** for releasing the NO-containing gas. Alternatively, it is envisioned that the tip of the device **10** may be manufactured from more or less components. In the illustrated embodiment, as will be described in greater detail below, the jacket assembly **210** may be in the form of a compression nut. In use, rotation of the nozzle tip driver **220** may advance or move the anode **20** (e.g., the nozzle tip **200**, the coolant divider **160**, and the proximal anode **150**) towards the interim electrode **120** and the cathode **60** until the anode **20** (e.g., proximal anode **150**) contacts a chamber standoff, spacer or cylinder (used interchangeably herein without the intent to limit) **400** disposed between the interim electrode **120** and the anode **20** (e.g., proximal anode **150**) to ensure a precise distance between the interim electrode **120** and the anode **20** (e.g., proximal anode **150**).

(34) As shown in FIGS. **2-4**, and as previously mentioned, the outer housing **30** may be formed from two components, a lower shell **32** and an upper shell **40**. The lower shell **32** may include a proximal end **34** and a distal end **36**. Similarly, the upper shell **40** may include a proximal end **42** and a distal end **44**. The distal end **36** of the outer housing **30** may include a plurality of threads **38** for engaging a corresponding plurality of threads **46** formed on the proximal end **42** of the upper shell **40**. As previously mentioned, the proximal end **34** of the outer housing **30** may also include a plurality of internal threads **39** for engaging external threads **106** formed on a proximal end **102** of the insulator **100**. In addition, the insulator **100** may include a plurality of internal threads **107** for engaging external threads **63** formed on the cathode **60** (e.g., proximal cathode **62**). In use, by threadably coupling the cathode **60** to the insulator **100**, a user is able to adjust the relative position of the cathode **60** (and hence the hafnium tip **80** coupled to a distal end thereof) with respect to the interim electrode **120**, which is coupled to the distal end of the insulator **100**, and with respect to the anode **20**, to achieve a desired NO concentration. For example, in use, a user can engage a proximal end of the cathode **60** via, for example, a tool such as, but not limited to, a screw driver, a socket drive, etc. and by rotating the cathode **60** adjust the position of the cathode **60** with respect to the anode **20** and the interim electrode **120**.

(35) In addition, the jacket assembly **210** may include a proximal end **212** and a distal end **214**. The distal end **44** of the upper shell **40** may include a plurality of threads **48** for engaging a corresponding plurality of threads **216** formed on the proximal end **212** of the jacket assembly **210**. Similarly, the nozzle tip driver **220** may include a plurality of threads **222** for engaging a corresponding plurality of threads **218** formed on the distal end **214** of the jacket assembly **210**. Tightening or loosening of the nozzle tip driver **220** when threaded in the jacket assembly **210** can allow for adjustable positioning of the anode **20** relative to the cathode **60** in order to produce various desired NO concentrations. It should be understood that while the lower shell **32**, the upper shell **40**, the external threads **106** formed on the insulator **100**, the jacket assembly **210** and the nozzle tip driver **220** have been described and illustrated as including a plurality of threads for

engaging one another, it is contemplated that the components may alternatively be coupled together by any other means now known or hereafter developed including, for example, being constructed as a single unitary component.

(36) As previously mentioned, the lower shell **32** may enclose the cathode **60** (e.g., proximal cathode **62**, distal cathode **64**, and hafnium tip **80**). In addition, the outer housing **30**, and more specifically, the lower shell **32**, may further enclose the interim electrode **120** and the insulator **100**. The insulator **100** may be made from a dielectric material, such as, for example, a polymer, a ceramic, PTFE (Polytetrafluoroethylene), etc. The insulator **100** may be positioned between the cathode **60** and the outer housing **30**, for example, between the cathode **60** and the lower shell **32** to insulate the cathode **60** and the lower shell **32** from one another. In addition, as previously mentioned, the insulator **100** also isolates the cathode **60** (e.g., the proximal and distal cathodes **62**, **64**) from the interim electrode **120**. The interim electrode **120** is used to initiate the arc and maintain/stabilize the continuity of electrical discharge from the cathode **60** to the anode **20**.

(37) The device **10**, and more specifically the lower shell **32**, may also include a cathode insulator **85** for providing a dielectric barrier between the outer housing **30** (e.g., lower shell **32**) and the cathode **60** for providing sufficient electrical insulation.

(38) The outer housing **30**, and more specifically, the upper shell **40**, may enclose the anode **20** such as the proximal anode **150**, the coolant divider **160** and the nozzle tip **200**. In use, when the high voltage supply is connected to the cathode **60**, an electrical arc is produced between the interim electrode **120** and the cathode **60** (e.g., hafnium tip **80**) and onto the anode **20** thus creating the NO-containing gas plasma in the interelectrode area **500** (e.g., area between the cathode **60** and the anode **20**) across the electric arc formed between the cathode **60** and the anode **20**. Thereafter, the NO-containing gas is directed through the output flow channels formed in the proximal anode **150**, the coolant divider **160**, and the nozzle tip **200**, all of which are in line with the cathode **60**. In this way, an output flow of the NO-containing gas is discharged through the nozzle tip **200**.

(39) As previously mentioned, the device **10** may include one or more pathways for receiving fluid therein. Referring to FIGS. **2** and **3**, in one embodiment, the device **10**, and more specifically the lower shell **32** and the insulator **100** may form first and second lower coolant pathways **310**, **312** coupled, respectively, to a fluid entry port **300** and a fluid exit port **302**. Similarly, the device **10**, and more specifically the upper shell **40** and the coolant divider **160** may form first and second upper coolant pathways **320**, **322** which are in fluid communication with the first and second lower coolant pathways **310**, **312**, respectively. In use, when the lower shell **32** is coupled to the upper shell **40**, the first and second lower coolant pathways **310**, **312** are aligned with the first and second upper coolant pathways **320**, **322**. In this manner, a fluid may be introduced into the device **10** via the fluid entry port **300**, circulated thru the device **10** via the first lower coolant pathway **310** and the first upper coolant pathway **320** located between the lower and upper shells **32**, **40** and the cathode **60** and anode **20**, respectively. At the distal end of the upper shell **40**, the cooling fluid may be passed to the second upper coolant pathway **322**, then to the second lower coolant pathway **312**, where it exits through the fluid exit port **302** located at the proximal end **34** of the outer housing **30**. More specifically, the cooling fluid may be introduced into the device **10** at the fluid entry port **300** located at the proximal end **34** of the lower shell **32**. The cooling fluid may travel the fluid pathways formed in the lower shell **32** and into the upper shell **40** adjacent to the coolant divider **160**, then back down and out of the device **10** via the fluid exit port **302** located at the proximal end **34** of the lower shell **32**.

(40) In general, as the cooling fluid travels through the coolant pathways formed within the outer housing **30**, heat from the NO-containing gas is transferred through the coolant divider **160** and into the cooling fluid, reducing the temperature of the NO-containing gas as it travels to the nozzle tip **200**.

(41) As will be described in greater detail below, the insulator **100** may also include one or more coolant entryways **108** for providing a pathway for the cooling fluid to interact with the cathode **60**.

That is, the insulator **100** may include one or more coolant entryways **108** for providing a pathway for the cooling fluid to directly contact at least a portion of the cathode **60** (e.g., distal cathode **64**) prior to the interim electrode **120**. The configuration of these coolant entryways **108** is designed to maximize thermal draw from the cathode **60** to the cooling fluid thereby reducing the operating temperature of the cathode **60** while under load.

(42) In one embodiment, the fluid could be electrically conductive. In contrast, dielectric fluid, such as ethylene glycol, propylene glycol, or silicone oil may be used to maximize thermal draw and to prevent the interim electrode **120** from achieving electrical potential from the cathode **60**.

(43) As illustrated, the insulator **100**, the coolant divider **160**, and other components located adjacent the coolant pathways **310**, **320**, **322**, **312** may have a substantially flat or ribbed shape or surface area to allow for increased cooling of the internal components. In this manner, the contact area is maximized between the insulator **100**, the coolant divider **160**, and the cooling fluid, thereby enhancing heat transfer (e.g., cooling efficiency) therebetween.

(44) In contrast with prior devices that provide cooling beyond the nozzle tip (i.e., a majority of the cooling of the NO-containing gas occurs after the NO-containing gas has exited the nozzle tip), the disclosed device **10** enables cooling, via the cooling fluid, to begin within the device **10** such as, for example, in the proximal anode **150**. That is, according to one aspect of the present disclosure, the cooling fluid may interact with and cool the NO-containing gas along a substantial length of the NO-containing gas flow outlet channel **152** between the proximal anode **150** to the nozzle tip **200**. In this manner, the exiting NO-containing gas is cooled to a much greater extent before it exits the device **10** as compared to prior devices. Due to the reduced temperature of the NO-containing gas exiting the nozzle tip **200**, the concentration of NO-containing gas that can be immediately directed to the treatment site, can be higher as compared to prior devices. In non-limiting exemplary embodiments, the NO-containing gas exiting the nozzle tip **200** may be approximately 50 degrees Celsius or less and may contain between 700 to 1,100 ppm of NO.

(45) As previously mentioned, the cathode **60** may further include a central hollow portion that provides a pathway **330** for air to be forced through the cathode **60** and into the interelectrode area **500** for use in generating a plasma in the interelectrode area **500**. Referring to FIGS. **9A**, **9B** and **9C**, in one embodiment, the distal end of the cathode **60** may include one or more tangential holes **600**. In use, the holes **600** transfer the air around the hafnium tip **80** and into the interelectrode area **500**. The tangential holes **600** may be configured to create a vortex in the airflow as the gas passes through the interelectrode area **500**. That is, in connection with the example embodiment of the cathode tip illustrated in FIGS. **9A**, **9B** and **9C**, as air comes in through the central hollow portion and out the holes **600**, the air swirls around the cathode **60** (e.g., hafnium tip **80**). The swirling airflow surrounds the plasma and its afterglow as it goes out through the interim electrode **120** and the anode **20**, which assists in arc stabilization, resulting in a more consistent NO production.

(46) In use, an air pump (not shown) may supply forced air through the proximal end **34** of the device **10** (via pathway **330**) into the cathode **60** and forced out the nozzle tip **200**. This allows the NO-containing gas created in the interelectrode area **500** to be forced up and through the nozzle tip **200** to allow it to be impinged on the treatment area.

(47) As previously mentioned, the interim electrode **120** may be coupled to the insulator **100**. Referring to FIGS. **5A** and **5B**, the insulator **100** may include a proximal end **102**, a distal end **104**, a plurality of threads **106** formed on the proximal end **102** for engaging threads **39** formed on the outer housing **30**, and a plurality of threads **110** (e.g., internal threads) formed on a distal end **104** thereof for threadably engaging threads **126** (e.g. external threads) formed on a proximal end **122** of the interim electrode **120**. In addition, as illustrated, the interim electrode **120** may include an enlarged flange **128** formed at a distal end **124** thereof so that the interim electrode **120** may be located in a fixed position relative to the insulator **100** and the cathode **60** when the threads **126** formed on the interim electrode **120** and the threads **110** formed on the insulator **100** are fully engaged. That is, by providing a flange **128** on the interim electrode **120** to butt against the distal

end **104** of the insulator **100**, a consistent distance between the interim electrode **120** and the cathode **60** can be maintained. It should be noted that while the insulator **100** and the interim electrode **120** have been illustrated and described as being threadably connected, it is envisioned that they may be coupled together by other means. In addition or alternatively, the device **10** may also incorporate an interim electrode standoff **90** disposed between the distal end of the distal cathode **64** adjacent the interim electrode **120** and the hafnium tip **80**. In one example embodiment, the interim electrode standoff **90** may be manufactured from a dielectric material. The interim electrode standoff **90** may be used in place of the flange **128** formed on the interim electrode to provide a precise gap distance between the distal cathode **64** and the interim electrode **120**.

(48) Referring to FIG. 5C, the insulator **100** may also include one or more geometric features **115** in fluid communication with the coolant entryways **108** for directly the coolant in a preferred direction to cool the interim electrode **120**. That is, as illustrated, the distal end **104** of the insulator **100** may include one or more geometric features **115** such as, for example, a recessed surface, shoulders, etc. to aid in directly the exiting fluid from the coolant entryways **108** over the interim electrode **120**. In this manner, an increase amount of fluid can be expected to exit the coolant entryways **108** and pass across or over the interim electrode **120**.

(49) The device **10** may further include one or more alignment features for interconnecting various components of the device **10** to facilitate assembly of the device **10**. For example, referring to FIGS. 6A-6C, the insulator **100** (FIG. 6A) may include one or more complementary features for coupling to the cathode insulator **85** (FIG. 6B). By providing complementary engagement features for coupling the insulator **100** to the cathode insulator **85**, the device **10** is better able to maintain a fixed, consistent alignment between the insulator **100** and the cathode insulator **85**. The complementary engagement features between the insulator **100** and the cathode insulator **85** may be any suitable features now know or hereafter developed. In one example embodiment, as illustrated, the insulator **100** may include one or more recesses **116** formed in, for example, the distal end **104** of the insulator **100**. Meanwhile, the cathode insulator **85** may include one or more inwardly protruding projections **88** formed on a distal end **87** thereof so that when the insulator **100** is inserted into an internal borehole **86** formed in the cathode insulator **85**, the projections **88** formed on the cathode insulator **85** mate with the recesses **116** formed in the insulator **100** and thus fix the position of the insulator **100** with respect to the cathode insulator **85**.

(50) Referring to FIGS. 7A and 7B, the cathode insulator **85** may also include one or more complementary features for coupling to the alignment bushing **450**, for example, to a proximal end **452** of the alignment bushing **450**. By providing complementary engagement features for coupling the cathode insulator **85** to the alignment bushing **450**, the device **10** is better able to maintain a fixed, consistent alignment between the cathode insulator **85** and the alignment bushing **450**. The complementary engagement features between the cathode insulator **85** and the alignment bushing **450** may be any suitable features now know or hereafter developed. In one example embodiment, as illustrated, the cathode insulator **85** may include one or more recesses **91** formed in, for example, the distal end **87** of the cathode insulator **85**. Meanwhile, the alignment bushing **450** may include one or more projections **454** formed on a proximal end **452** thereof so that when the cathode insulator **85** is coupled to the alignment bushing **450**, the projections **454** formed on the alignment bushing **450** mate with the recesses **91** formed in the cathode insulator **85** and thus fix the position of the cathode insulator **85** with respect to the alignment bushing **450**.

(51) Referring to FIGS. 8A and 8B, the alignment bushing **450** may also include one or more complementary features for coupling to the coolant divider **160**. By providing complementary engagement features for coupling the alignment bushing **450** to the coolant divider **160**, the device **10** is better able to maintain a fixed, consistent alignment between the alignment bushing **450** and the coolant divider **160**, and thus consistent alignment of the proximal anode **150**, which is coupled to the coolant divider **160**, and the interim electrode **120** and the cathode **60**. The complementary engagement features between the alignment bushing **450** and the coolant divider **160** may be any

suitable features now known or hereafter developed. In one example embodiment, as illustrated, the coolant divider **160** may include one or more recesses **164** formed in, for example, the proximal end **162** thereof. Meanwhile, the alignment bushing **450** may include one or more inwardly protruding projections **456** formed on the proximal end **452** thereof so that when the coolant divider **160** is inserted into an internal borehole **458** formed in the alignment bushing **450**, the projections **456** formed on the alignment bushing **450** mate with the recesses **164** formed in the coolant divider **160** and thus fix the position of the coolant divider **160** with respect to the alignment bushing **450**. The coolant divider **160** may also include one or more stop members **166** formed thereon for providing a limit stop for inserting the coolant divider **160** into the alignment bushing **450** to provide proper spacing between the coolant divider **160** and the alignment bushing **450**, and thus providing proper spacing between the proximal anode **150** and the interim electrode **120**.

(52) By manufacturing and assembling a device in accordance with the features described herein, it is contemplated that proper and consistent alignment and positioning of the various components including, for example, the distal cathode **64**, the interim electrode **120**, and the proximal anode **150** can be achieved when the threads **126** formed on the interim electrode **120** mate with the threads **110** formed on the insulator **100** so that the flange **128** of the interim electrode **120** contacts the distal end **104** of the insulator **100**, and when after proper positioning of the various components, the jacket assembly **210** is secured onto the distal end **44** of the upper shell **40** and the nozzle tip driver **220** is threaded into engagement with the threads formed on the jacket assembly **210** so that rotation of the nozzle tip driver **220** advances the nozzle tip **200**, the coolant divider **160**, and the proximal anode **150** towards the cathode **60**. It is also contemplated that rotation of the nozzle tip driver **220** enables adjustment of the distance between the anode **20** and the cathode **60** for varying the NO concentration.

(53) In addition, the incorporation of complementary engagement features to couple the insulator **100** with the cathode insulator **85**, to couple the alignment bushing **450** with the cathode insulator **85**, and to couple the alignment bushing **450** with the coolant divider **160** helps to facilitate internal alignment of the various components of the device **10**, which, in turn, assists with ensuring proper assembly of the device **10** and allows for adjustment of the spacing between the interim electrode **120**, the cathode **60**, and the anode **20**.

(54) Referring to FIGS. 2-4, the device **10** may further include a chamber standoff or spacer **400** disposed between the interim electrode **120** and the proximal anode **150**. The chamber standoff **400** may be associated with a chamber seal **402**, for example, the chamber seal **402** may be in the form of a seal for externally surrounding the chamber standoff **400**, the chamber seal **402** sealing the chamber area between the interim electrode **120** and the proximal anode **150**. In the illustrated embodiment, the chamber standoff **400** is adjacent to the distal end **36** of the lower shell **32** and the proximal end **42** of the upper shell **40**. The chamber standoff **400** may, in some embodiments, simplify manufacturing and assembly of the device **10**. That is, in combination with the nozzle tip driver **200**, the chamber standoff **400** may be used to ensure a precise distance between the interim electrode **120** and the proximal anode **150** is achieved regardless of the manufacturing tolerances of the other elements of the device **10**. When assembled, rotation of the nozzle tip driver **220** compresses the nozzle tip **200**, the coolant divider **160**, and the proximal anode **150** towards the cathode **60** until the chamber standoff **400**, which is in contact with the proximal anode **150**, contacts the distal surface **124** of the interim electrode **120** to provide an exact gap distance between the anode **20** and the cathode **60**. As will be appreciated, providing a precisely repeatable standoff between the anode **20** and the cathode **60** can be important because it provides for a consistent arc gap thereby maintaining unit to unit output parameters. In one embodiment, the chamber standoff **400** may be manufactured from a ceramic material, although other suitable materials are envisioned.

(55) In use, the nozzle tip driver **220** allows torqueing of the internal components (e.g., the nozzle

tip **200**, the coolant divider **160**, the proximal anode **150**, etc.) down onto the distal end of the chamber standoff **400**. As mentioned, regardless of the manufacturing tolerances of these elements, by providing a highly tolerance chamber standoff **400**, a high-tolerance, highly accurate standoff or spacing between the interim electrode **120** and the proximal anode **150** may be achieved. As such, the chamber standoff **400** provides a single high tolerant component, minimizing the need for the remaining components to be individually adjusted, thus greatly increasing the manufacturability, repeatability and verifiability of the device **10** simply by torqueing the nozzle tip driver **220** down onto the chamber standoff **400**.

(56) The device **10** may also include a plurality of O-rings **475** and machined grooves between components to maintain appropriate fluid seals between internal components to prevent the cooling fluid from entering the air and electrical sections (and vice versa). For example, an O-ring may be positioned between the jacket assembly **210** and the distal end **44** of the upper shell **40**; between the nozzle tip **200** and the coolant divider **160**; between the distal end **36** of the lower shell **32** and the proximal end **42** of the upper shell **40**; between the distal cathode **64** and the insulator **100**; etc.

(57) As previously mentioned, in use, adjustment of NO concentration can be achieved by adjusting either or both of (i) the distance between the anode **20** and the cathode **60** (e.g., the relative distance between the anode **20** and the cathode **60** (hafnium tip **80**), and/or (ii) the voltage applied to the cathode **60** (e.g., hafnium tip **80**). As will be understood, by varying the offset or distance (e.g., distance X in FIG. 3A) between the cathode **60** and the anode **20**, the amount of voltage needed to be applied to the cathode **60** to maintain the plasma arc is varied, and thus the NO concentration in the gas is also varied.

(58) In use, the distance or relative position between the anode **20** and the cathode **60** can be achieved by any mechanism now known or hereafter developed. In one embodiment, as previously mentioned, the device may include a chamber standoff or spacer **400** disposed between the interim electrode **120** and the proximal anode **150**. The chamber standoff or spacer **400** is used to provide a known, repeatable offset or distance between the anode **20** and the cathode **60**. In one embodiment, the device may be provided with a plurality or set of chamber standoffs or spacers **400** having various dimensions (e.g., different widths). In use, each width would correspond to a different offset between the anode **20** and the cathode **60**, and a different voltage applied to the cathode **60**. The result is that each chamber standoff or spacer **400** could provide a predetermined NO concentration in the resulting gas. The various different chamber standoffs or spacers **400** providing a different concentration. As such, to adjust the NO concentration to a desired level, the user needs only to insert the corresponding chamber standoff or spacer **400** into the device and assemble the device with the anode **20** in contact with the chamber standoff or spacer **400**, as previously described. Thereafter, the user tightens the nozzle tip driver **220** until the anode **20** is secured in place. The desired offset or distance between the anode **20** and the cathode **60** is automatically achieved based on the selected chamber standoff or spacer **400**. For example, in one embodiment, by providing a cathode **60** to anode **20** distance of 5 mm, via for example, an appropriately sized chamber standoff or spacer **400**, with 230V at 1.8 amps applied to the cathode, produces approximately 1000 ppm NO in the resulting gas stream. It will be appreciated that this example is not limiting, and that other standoff/spacer dimensions, applied voltages/currents and air flow rates can be used to adjust the concentration of NO in the resulting gas, as well as the concentration of NO in the resulting gas stream.

(59) In an alternate embodiment, as previously mentioned, the distance between the anode **20** (e.g., proximal anode **150**) and the cathode **60** (e.g., hafnium tip **80**) may be adjusted by rotating the cathode **60** (e.g., proximal cathode **62**) with respect to the insulator **100** using, for example, a tool such as, a screwdriver, a drive socket, etc. coupled to an appropriate recess at the proximal end of the cathode **60**. In use, rotation of the cathode **60** with respect to the insulator **100** causes the mating threads to advance or retract, depending on the direction of rotation, the cathode **60** (e.g., hafnium tip **80**) with respect to the anode **20** (e.g., proximal anode **150**). Similar voltages would be

applied (depending on the magnitude of the resulting offset) and resulting ranges of NO concentrations would be obtained. For example, one could fine tune the operation of the device by selecting an appropriately sized chamber standoff or spacer **400**, then make additional, small offset adjustments using the threaded connection between the cathode **60** and the insulator **100** to slightly advance or retract the cathode **60** (e.g., hafnium tip **80**).

(60) In yet another embodiment, the distance between the anode **20** and the cathode **60** can be adjusted using an adjustable anode **800**. Referring to FIGS. **10A** and **10B**, the adjustable anode **800** may include inner and outer components **820**, **840**, for example, an inner electrically conductive component **820** and an outer component **840**. In use, the adjustable anode **800** may take the place of the anode **20** located within the outer housing **30** of the device. The inner component **820** may be functionally identical to anode **20** previously described, and may be made as a single component (as illustrated in FIGS. **10A** and **10B**) or from multiple components, as previously described. In use, the inner electrically conductive component **820** may be movably adjustable with respect to the outer component **840**. For example, as illustrated, the inner electrically conductive component **820** may include a plurality of external threads **822** for engaging internal threads **842** formed on an inner surface of the outer component **840**. As such, the distance between the anode and the cathode may be adjusted by rotating the inner component **820** of the anode with respect to the outer component **840** of the anode using, for example, a tool such as, a wrench, etc. In use, rotation of the inner component **820** of the anode with respect to the outer component **840** of the anode causes the mating threads to advance or retract, depending on the direction of rotation, the anode with respect to the cathode. Similar voltages would be applied (depending on the magnitude of the resulting offset) and resulting ranges NO concentrations would be obtained.

(61) In yet another embodiment, the position of the cathode **60** can be moved (e.g., extended or retracted) with respect to the anode **20** by rotating the insulator **100** relative to the outer housing **30** (e.g., the lower shell **32**) of the device. By adjusting the position of the insulator **100** relative to the outer housing **30**, one can adjust the position of the cathode **60** relative to the anode **20**.

(62) As used herein, an element or step recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “one embodiment” of the present invention are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(63) While certain embodiments of the disclosure have been described herein, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. A method for forming NO-containing gas flow to treat a biological object, the method comprising: forming an NO-containing gas flow in an interelectrode area of a device, wherein the device comprises: an anode; a cathode; an insulator disposed about the cathode; a cathode insulator disposed about a portion of the insulator; the interelectrode area disposed between the cathode and the anode; an interim electrode disposed about a proximal end of the cathode; a chamber standoff coupled between the interim electrode and the anode; wherein the insulator, cathode insulator, and chamber standoff are made from a dielectric material; directing the NO-containing gas flow from the device via an NO-containing gas flow outlet channel leading from the interelectrode area to a nozzle; and directing the NO-containing gas flow from the nozzle toward the biological object to treat the biological object.

2. The method of claim 1, further comprising creating a vortex of fluid flow as the fluid flow passes through the interelectrode area using plurality of holes formed in an outer perimeter of a distal end of the cathode.
 3. The method of claim 2, wherein the plurality of holes comprise a plurality of tangentially oriented holes.
 4. The method of claim 2, wherein the plurality of holes are configured to direct fluid from a central hollow portion of the cathode out of the plurality of holes such that the fluid swirls around the cathode to thereby surround a plasma and its afterglow as it goes out through the interim electrode and the anode to thereby assist in in arc stabilization.
 5. The method of claim 2, further comprising further comprising enabling an electrical arc to occur between the interim electrode and a hafnium tip associated with the cathode, the electrical arc emanating into the interelectrode area.
 6. The method of claim 1, further comprising providing a pathway for a cooling fluid to interact with the cathode using one or more coolant entryways in the insulator.
 7. The method of claim 1, further comprising providing a pathway for air to be forced through the cathode and into the interelectrode area using a central hollow portion of the cathode.
 8. The method of claim 1, further comprising separating the NO-containing gas flow outlet channel from a cooling fluid while directing the NO-containing gas flow to the nozzle using a fluid divider having first and second flat surfaces.
 9. A method for forming NO-containing gas flow to treat a biological object, the method comprising: forming an NO-containing gas flow in an interelectrode area of a device, wherein the device comprises: an anode, a cathode, and an insulator surrounding the cathode, the cathode including a plurality of tangentially arranged holes for creating a vortex of a fluid flow as the fluid exits the cathode; a cathode insulator disposed about the insulator; the interelectrode area between the cathode and the anode; an interim electrode disposed about a proximal end of the cathode; an interim electrode standoff coupled between the insulator and the interim electrode; and a chamber standoff coupled between the interim electrode and the anode; directing the NO-containing gas flow from the device via a flow outlet channel leading from the interelectrode area; and directing the NO-containing gas flow from the nozzle toward the biological object to treat the biological object.
 10. The method of claim 9, wherein at least one of the insulator, cathode insulator, and chamber standoff are made from a dielectric material.
 11. The method of claim 9, further comprising creating said vortex of fluid flow as the fluid flow passes through the interelectrode area using the plurality of tangentially arranged holes.
 12. The method of claim 9, wherein the plurality of holes are configured to direct a fluid from a central hollow portion of the cathode out of the plurality of holes such that the fluid swirls around the cathode to thereby surrounds the plasma and its afterglow as it goes out through the interim electrode and the anode to thereby assist in in arc stabilization.
 13. The method of claim 9, further comprising enabling an electrical arc to occur between the interim electrode and a hafnium tip associated with the cathode, the electrical arc emanating into the interelectrode area.
 14. The method of claim 9, further comprising providing a pathway for a cooling fluid to interact with the cathode using one or more coolant entryways in the insulator.
 15. The method of claim 9, further comprising providing a pathway for air to be forced through the cathode and into the interelectrode area using a central hollow portion of the cathode.
 16. The method of claim 9, further comprising separating the NO-containing gas flow outlet channel from a cooling fluid while directing the NO-containing gas to the nozzle using a fluid divider having first and second flat surfaces.
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